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THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Kaprekar Contest

(NMTC-Maths Olympiad at **SUB-JUNIOR LEVEL** *VII to VIII Standards)*

Saturday, 26th August 2006.

Note:

1) *Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.*
2) *Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
3) *For each correct response you get 1 mark;* ***for each incorrect response you lose ¼ mark.***
4) *Diagrams are only visual aids; they are not drawn to scale.*
5) *You are free to do rough work on separate sheets.*
6) *Duration of the test: 2 p.m. to 4. p.m – 2hours.*

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1. Given the alphametic where each letter represents a different digit, the value of C in the least value for ABCD is

$$\begin{array}{r} ABCD \\ -\ BCDA \\ \hline 1\,5\,1\,2 \end{array}$$

(A) 1 **(B) 2**
(C) 6 **(D) 7**

2. If $36a^4 = a^6$, then a^3 is equal to

(A) $\frac{1}{6}a^6$ **(B)** $6a^4$ **(C)** $\frac{1}{6}a^2$ **(D)** $6a^2$

3. The weight of a dog is 8 kg plus one third its weight. The weight of the dog is

(A) 11 kg. **(B) 12 kg.** **(C) 14 kg.** **(D) 15 kg.**

4. Sum of all integers less than 100 which leave a remainder 1 when divided by 3 and leave a remainder 2 when divided by 4 is

(A) 416 **(B) 1717** **(C) 1250** **(D) 1314**

5. Let n be a 3 digit number such that n = sum of the squares of the digits of n. The number of such n is

(A) 0 **(B) 1** **(C) 2** **(D) more than two**

6. A three digit number with digits A, B, C in that order is divisible by 9. A is an odd digit and C is an even digit. B and C are non zero. The number of such three digit numbers is

(A) 4 **(B) 8** **(C) 16** **(D) 20**

7. The number of prime numbers less than 1 lakh, whose digital sum is 2 (digital sum of a number is the sum of its digits) is,
(A) 5 (B) 4 (C) 3 (D) none of these

8. The quotient of 100^{100} and 50^{50} is
(A) 50^{50} (B) 50^{100} (C) 200^{50} (D) 400^{50}

9. A sequence of numbers $T_1, T_2, T_3, \ldots, T_n, \ldots$ is constructed using the rule that the n^{th} term T_n has n n's; For example the 25^{th} term $T_{25} = 25252525\ldots.25$, where 25 is repeated 25 times. The term containing the same number of 2's as T_{226} is
(A) T_{622} (B) T_{262} (C) T_{254} (D) T_{452}

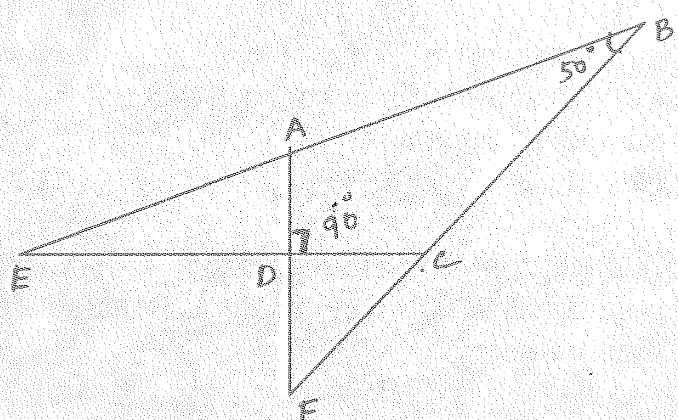
10. The least number of numbers to be deleted from the set {1, 2, 3,...,13, 14, 15} so that the product of the remaining numbers is a perfect square is
(A) 1 (B) 2 (C) 3 (D) 4

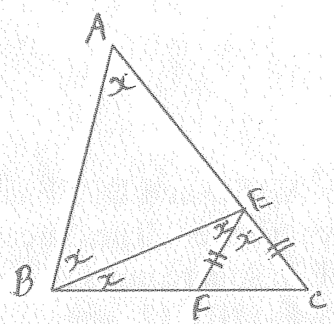
11. The digit 1 is attached to the right of a 3 digit number making it a 4 digit number which is 7777 more than the given number. The sum of the digits of the number is
(A) 23 (B) 18 (C) 17 (D) 16

12. A number is formed by writing the first 10 primes in the increasing order. Half of the digits are now crossed out, so that the number formed by the remaining digits without changing the order, is as large as possible. The second digit from the left of the new number is
(A) 2 (B) 3 (C) 5 (D) 7

13. Nine numbers are written in ascending order. The middle number is also the average of the nine numbers. The average of the 5 largest numbers is 68 and the average of the 5 smallest numbers is 44. The sum of all the numbers is
(A) 540 (B) 450 (C) 504 (D) 501

14. In the adjacent figure BA and BC are produced to meet CD and AD produced in E and F. Then $\angle AED + \angle CFD$ is
(A) 80° (B) 50°
(C) 40° (D) 160°



15. 

In a triangle ABC, BE is the angle bisector of $\angle ABC$, where E lies on AC. EF is the angle bisector of $\angle BEC$, where F lies on BC. Also EF = EC. Then,

(A) AB = BC **(B)** AC = BC
(C) $\angle ABC = \angle ACB = 72^\circ$ **(D)** $\angle BAC = 72^\circ$

16. In a box there are some coins and rings which are made of either gold or silver. 60% of the objects are coins, 40% of the rings are gold and 30% of the coins are silver. The percentage of gold articles is

(A) 24% **(B)** 16% **(C)** 42% **(D)** 58%

17. In a plane 3 lines and a circle are given. If points of intersection of two lines or that of a line with the circle are counted, the maximum number of points of intersection possible in this is

(A) 12 **(B)** 9 **(C)** 6 **(D)** 5

18. In an isosceles acute angled triangle one angle is 50°.

I. The other two angles are 65° and 65°.
II. The other two angles are 50° and 80°
III. The angles can be anything as long as the triangle is isosceles.

Then which of the following statements is true?

(A) I only **(B) II only** **(C) I and II only** **(D) III only**

19. Let n be the number of integers less than 10,000 which are divisible by all integers from 2 to 10. Then

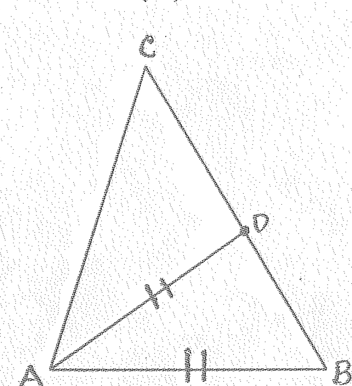
(A) $n = 0$ **(B)** $1 \leq n < 5$ **(C)** $5 \leq n < 10$ **(D)** $10 \leq n < 15$

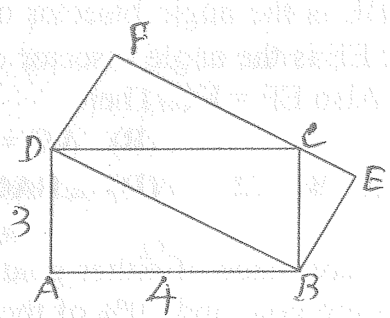
20. In an isosceles triangle the equal sides are 7 units each and the length of the base is an integer. From these a triangle with the greatest perimeter is selected. Its perimeter is

(A) 23 **(B)** 25 **(C)** 27 **(D)** 29

21. In an isosceles triangle ABC, AC = BC, $\angle BAC$ is bisected by AD where D lies on BC. It is found that AD = AB. Then $\angle ACB$ equals

(A) 72° **(B)** 54°
(C) 36° **(D) none of these**



22. 

Two rectangles ABCD and DBEF are as shown in the figure. The area of rectangle DBEF in square units is

(A) 10 (B) 12

(C) 14 (D) 15

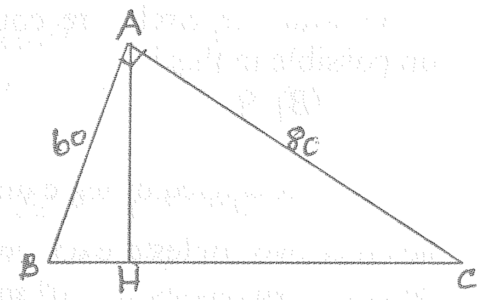
23. If the average of 20 different positive numbers is 20 then the greatest possible number among these 20 numbers can be

(A) 210 (B) 200 (C) 190 (D) 180

24. ABC is a right angled triangle with $\angle BAC = 90^{o}$. AH is drawn perpendicular to BC where H lies on BC. If AB = 60 and AC = 80, then BH =

(A) 36 (B) 32

(C) 24 (D) 30



25. Let the length of a positive integer n represent the number of prime factors of n, counting repetitions. Length of 36 is 4 as 36 = 2. 2. 3. 3. Length of 64 is 6 as $64 = 2^6$. The number of numbers less than 100 with maximum length is

(A) 5 (B) 4 (C) 3 (D) 2